

Faculty of Computer Science
Data Science and Business Analytics

Research seminar «Data analysis in finance»

Price Impact or Trading Volume: Why Is the Amihud (2002) Measure Priced?

X. Lou, T. Shu, *Review of Financial Studies*, 2017 · replication on Binance

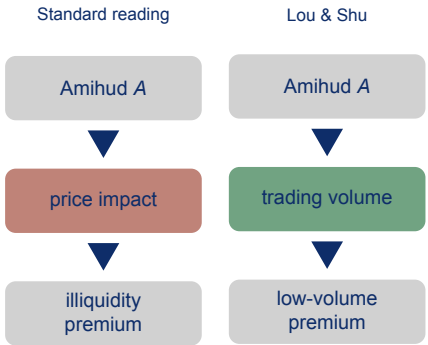
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Problem

Illiquid stocks earn a premium.
Is it for price impact, or just for low volume?

$$A = \frac{1}{D} \sum_{d} \frac{|r_d|}{\mathrm{DVol}_d}$$



Amihud is almost only volume

0.899

corr(A, A_C) · US stocks, paper

0.956

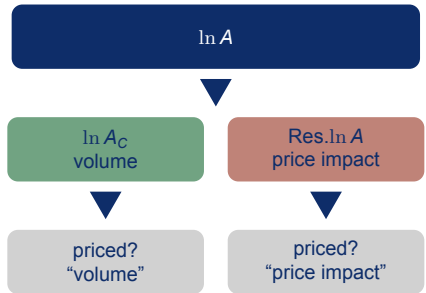
corr(A, A_C) · Binance, us

A_C keeps only $1/\mathrm{DVol}$, and it explains almost all of A.

Method: split Amihud in two

$$\ln A = a_t + b_t \ln A_C + \text{Res.} \ln A$$

If only the volume part is priced, the premium is about volume.



Test design



month $t-2$
MEASURE

month $t-1$
SKIP

month t
RETURN

$$r_{i,t} = \gamma_0 + \gamma_1 x_{i,t-2} + \text{controls} + \varepsilon_{i,t}$$

Quintile sorts and Fama–MacBeth, as in the paper.

The paper's result

H-L on A	H-L on A_C	H-L on Res.A
0.56 (t 2.36)	0.61 (t 2.95)	−0.17 (t −1.05)
%/month	%/month	%/month

Volume is priced, the rest is not: the premium is not for price impact.

First, simulations

World V
volume is priced

100%

runs find A_C priced

World P
price impact is priced

100%

runs find Res.$\ln A$ priced

World 0
nothing is priced

9.5%

false positives (target 5%)

The pipeline tells the hypotheses apart before we touch real data.

Dataset overview



Binance spot · USDT pairs

626 coins

93 months

24 628 coin-months

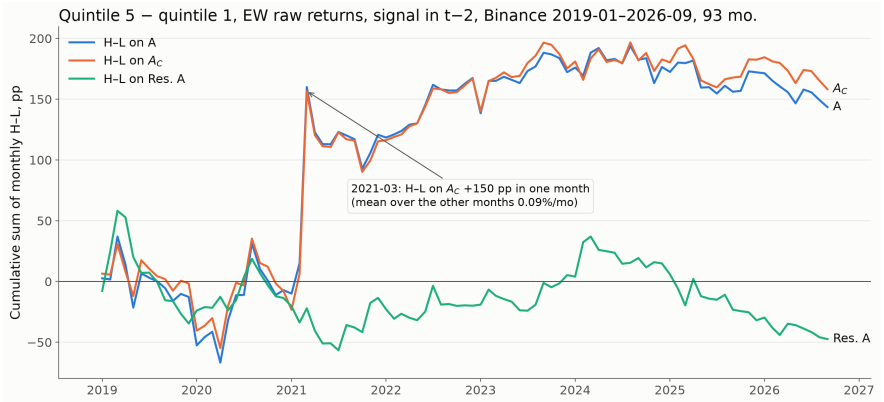
3 signals: A, A_C, Res.A

Data analysis: what differs from the paper

	paper	Binance
spread of A_C, Q75/Q25	116.60	9.40
corr(A, $\lvert Ret \rvert$)	0.489	−0.133
s.d. of monthly return	—	43.9%

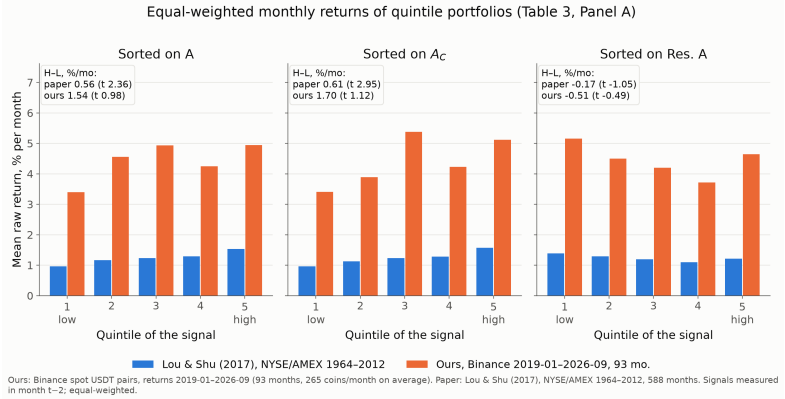
Coins differ less in liquidity, move much more, and trade more when volatile.

One outlier month: March 2021



The altcoin season makes the volume spread. Without it: 0.09% per month.

Our replication next to the paper



Same signs, but noisy: H-L on A_C 1.70 (t 1.12).

Upgrades

0.615
corr with 5-min λ
(paper 0.737)

98.8%
months with COV < 0
(Amihud & Noh premise)

2.57 (t 1.84)
sentiment long-short

0.308 (t 2.87)
$\ln A_C$ on MOEX

Amihud tracks true price impact, but no upgrade settles the debate.

The paper versus us

25/28
signs agree

8/21
significant results repeat

7.0×
smallest premium we could detect, vs the paper's

The direction replicates. With 93 months instead of 588, significance does not.

Overfitting

16/73
tests pass 1.96
(3.65–5.7 expected by chance)

0
survive Bonferroni

2/11
stable in the bootstrap

Every test logged: our evidence is weak, and we say so.

Takeaways

Amihud is mostly volume.

The paper’s signs replicate on crypto.

Its significance needs more than 93 months.

Price Impact or Trading Volume (Lou, Shu, 2017)

